

Multi-Beam Direct-Write Electron Lithography

A Plain-Language Guide to the Morin 2026 v5 Open Architecture

A 30-minute walkthrough for bright non-specialists

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About this document. This is a plain-language companion to the technical preprint *Multi-Beam Direct-Write Electron Lithography via Multi-Rate Electromagnetic Steering* (Morin 2026, v5). The technical preprint is 24 dense pages of physics, engineering specifications, and cost analysis. This companion document covers the same ground in roughly 15 pages, using analogies and concrete numbers, for readers with a STEM background but no semiconductor specialty. Released under the same CC BY 4.0 license as the parent work. No patent claim is made on anything described herein.

What this is, in one paragraph

Imagine a million tiny robot monks, each with its own quill, all copying the same intricate page in perfect synchrony onto a silicon canvas. That, in a sentence, is what this machine does — except the “quills” are electron beams, the “page” is the wiring pattern of a microchip, and the “monks” are micro-scale electromagnetic coils that have been frozen to 196 degrees below zero so they can work without melting. The whole apparatus is the size of a refrigerator, costs about \$37 million, draws its instructions through optical fibres, and is fully open-source: no patents, no single-vendor lock-in, no country with a monopoly on the design. The architecture has been simulated end-to-end, every subsystem has at least three independent suppliers across at least two countries, and the first physical prototype is buildable today for about \$600,000 over 18 months. It is one credible answer to a question that has quietly become a problem of geopolitics: *who is allowed to make advanced computer chips, and at whose pleasure?*

1 The problem

1.1 The world according to ASML

If you want to make a leading-edge computer chip today — the kind that goes inside an iPhone or a recent NVIDIA GPU — there is exactly one company on Earth that sells the machine you need. The company is ASML, based in the Netherlands. The machine is called an EUV scanner. It costs roughly \$200 to \$400 million per unit, weighs as much as a Boeing 737, and ships in dozens of shipping containers. ASML’s manufacturing lead time runs to years. Each scanner is, individually, the most expensive industrial machine in mass production.

The word “lithography” simply means *writing patterns very precisely onto a surface*. Chip lithography means writing the wiring diagram of a computer chip onto a polished silicon wafer, where

the wires are only a few atoms wide. EUV stands for *extreme ultraviolet*: light at a wavelength so short that there is no glass that can focus it, no air it can travel through, and almost no mirror that can reflect it. ASML solved this by using mirrors made of dozens of nanometre-thin layers, in a vacuum, with the light generated by smashing droplets of tin with a laser fifty thousand times per second. It works. It works beautifully. It is also the only thing on Earth that works for the cutting edge.

This creates several problems at once.

The price is exclusionary. A \$400 million machine prices out everybody except the half-dozen most heavily capitalised chipmakers on the planet. Universities cannot buy one. Defence labs cannot easily buy one. Most countries cannot buy one. The total number of EUV scanners that have ever been delivered, anywhere, is in the low hundreds.

The mask cost is exclusionary too. EUV doesn't just print directly onto a wafer. It uses a *mask* — think of a stencil, or more accurately a movie reel — that the light passes through. A mask for a single chip layer costs \$300,000 to \$1 million. A full chip needs dozens of mask layers. The total mask cost for one new chip design easily exceeds \$50 million. If you're TSMC making a billion iPhones, this is fine. If you're a startup trying to make a thousand specialty chips for a satellite, it is not.

The supply chain is single-point. ASML's most critical sub-supplier, the light-source company Cymer, is also effectively unique. The mirrors come from one company in Germany. The export of EUV scanners is controlled by the Dutch and U.S. governments and has become, openly, a tool of geopolitics. If ASML's factory in Veldhoven were destroyed tomorrow, the global rate of cutting-edge chip production would simply stop.

This is the situation at the high end. Now look at the other extreme.

1.2 The monk with a quill

Long before ASML existed, scientists made tiny patterns by aiming a single electron beam at a wafer and steering it around, one pixel at a time, like a person carefully drawing with a very fine pen. This technique is called single-beam electron-beam lithography, or single-beam EBL.

A single electron beam can write features down to a few nanometres — finer than EUV in fact. It needs no mask: you simply tell the beam where to go. It is, in principle, the perfect tool for low-volume custom chips.

The problem is speed. One beam, drawing one pixel at a time, has to sweep across the entire wafer. A modern chip might have a trillion pixels per layer. Writing one wafer can take *days*. A high-volume chip fab cannot operate on day-per-wafer economics; it needs to push a wafer through every few minutes. So single-beam EBL has been relegated, since the 1970s, to making the very masks that EUV machines use — and to research labs and university clean rooms where time is cheaper than money.

This is the monk with the quill. He writes beautifully. He is uselessly slow.

1.3 What's been missing

For decades there has been an obvious gap in the market: a middle path. A machine that writes patterns directly (no mask), in parallel (many beams at once, so it's fast), at moderate resolution

(not bleeding edge, but good enough for the 90% of chips that don't need to be cutting-edge), at a moderate price (millions, not hundreds of millions), with multiple suppliers for every part (so no one company or country can hold it hostage).

Several companies have built fragments of this — IMS Nano in Austria makes a multi-beam mask writer; Mapper Lithography in the Netherlands tried to make a direct-write tool and went bankrupt in 2018. None of them have produced an open, mature-node, mask-free direct-write tool that anyone can buy or build.

That is the gap this architecture fills.

2 The big idea

The big idea has three moves, each of which sounds simple and each of which took a generation of physicists and engineers to make possible.

2.1 Move one: split the monk into a million monks

Instead of one electron beam tracing the entire pattern, use a million electron beams in parallel, each responsible for its own little tile of the wafer. A million beams running together can write a wafer in a few hours instead of a few days. This is straightforwardly the difference between a single laser scanning back and forth across a screen, and a holographic projector throwing the entire image at once.

But each beam needs to be aimed independently. The pattern on your chip is not symmetrical — each beam writes a different shape. So each beam needs its own steering mechanism. In practice, this means each beam needs its own tiny electromagnetic coil — a “Lorentz coil” — that bends the beam by a small, precisely-controlled amount, in two axes. Call this the steering wheel for each monk.

A million steering wheels in a column the size of a soda can. Each one switchable thousands of times per second. Each one accurate to a few parts per million.

2.2 Move two: the toaster in a thermos

Here is where things stop sounding simple. Each of those million coils is a tiny copper microcoil. When you pass current through copper, you get heat. A million microcoils, switching ten thousand times a second, dissipates about one kilowatt of heat. That is roughly the same as a kitchen toaster.

The toaster is inside the electron column. The electron column has to be in ultra-high vacuum — better than the vacuum of low Earth orbit — because electrons are scattered by even a single gas molecule. *There is no air in there.* That means there is no convection, no air currents carrying heat away. The only way heat can leave is by very slow infrared radiation.

You have just put a kilowatt-toaster inside a thermos. The column will heat up by tens of degrees per second. Within a fraction of a second, the coils will deform, the alignments will drift, and the entire instrument will be useless. Within seconds, the coils themselves will melt.

This was the showstopper that killed several earlier multi-beam direct-write attempts. You cannot just air-cool a vacuum.

2.3 Move three: the cold-room rescue

The solution is to take the toaster out of the room-temperature thermos and put it instead in a deep freezer. Specifically: cool the entire electron column to 77 kelvins. That is 77 degrees above absolute zero, or about -196°C — the temperature of liquid nitrogen. Cold enough to instantly freeze a banana so hard it shatters like glass.

Two things happen at 77 K that solve every part of the heat problem.

First, the coils become superconducting. Made of a material called YBCO (yttrium-barium-copper-oxide), the coils have *zero* electrical resistance below about 90 K. Zero resistance means zero heat from current flowing through them. The toaster turns off. Completely.

Second, the coils become a magnetic mirror. YBCO is a high-temperature superconductor, and superconductors do something called the Meissner effect — they expel magnetic fields entirely. A thin shield of YBCO between two coils means that, magnetically, neither coil can “see” the other. They cannot interfere with each other. The crosstalk between beams that would otherwise destroy alignment simply vanishes. It’s like putting a perfect mirror between two flashlights pointed at each other.

You still have to cool the column down to 77 K and keep it there. That is the job of a *cryocooler*, which is essentially an industrial-strength version of the heat-pump cooling cycle in your refrigerator. Cryocoolers in this class are sold by half a dozen companies (Cryomech, Sumitomo, Brooks) for \$30,000 to \$80,000 each. They are commodity equipment, used routinely in MRI machines, satellite test chambers, and quantum computing labs. No monopoly. No long lead times. No export controls.

The remaining heat from cryo-CMOS electronics — the chips that control the coils — is a manageable 150 watts, and the design uses two cryocoolers in parallel for redundancy. Total cooling capacity: 250 watts at 77 K, with comfortable engineering margin.

2.4 Move four (which falls out for free): no mask

Because every beam is independently steered, *anything you can tell the beams to do, they can do*. Want to write a different pattern next week? Send different instructions; no mask change. Want to write a custom one-off prototype chip? Just send the new pattern. Want to make small adjustments mid-batch to compensate for tool drift? Update the instructions in real time.

This is the disruptive part of the disruption. EUV needs a mask, which means every new chip design has a \$50-million entry fee. This machine has zero mask cost, ever. The marginal cost of trying a new design is the cost of a few wafers and a few hours of tool time.

3 The architecture explained

Let’s walk through the physical machine, top to bottom.

3.1 The column: where the electrons are born and steered

At the top of the machine, behind several layers of vacuum, sits the *source array*: a tiny chip with a million sharp pointy tips on it, each one a carbon nanotube or a Spindt-type metal tip, each one emitting a stream of electrons when a high voltage is applied. These tips are spaced 20

micrometres apart — about a quarter the width of a human hair between neighbours. A million tips fit comfortably onto a chip 20 millimetres on a side.

Below the tip array, each electron stream is accelerated by 50,000 volts. At 50 kilovolts, an electron is moving at about 40% of the speed of light — fast enough that a packet of electrons leaves the tip and hits the wafer in less than a billionth of a second. This is important: you can switch the pattern many thousand times per second and the beams keep up.

Then comes the deflection section: those million tiny YBCO superconducting coils, one per beam, doing the steering. Each coil bends its beam by a small angle, deflecting it across a tile of wafer roughly 20 micrometres on a side. Each beam covers its own tile. Together, the million beams cover a 20-millimetre-square area in a single sweep.

Below the deflection coils sits the *blanker array* — a layer of tiny electrostatic plates, one pair per beam. The blanker is the on/off switch for each beam. When a beam needs to write, the blanker lets it through. When the beam needs to not-write, the blanker deflects it sideways into a graphite stop. The blanker switches at gigahertz speeds — billions of times per second.

The whole column — source array, accelerator, deflection coils, blankers — is the size and shape of a stubby coffee thermos, all maintained at 77 K, all in ultra-high vacuum, all inside the cryocooler shroud.

3.2 The wafer: the canvas

Below the column sits the wafer: a polished disc of silicon, 300 millimetres (about 12 inches) across, ground flatter than the Earth’s curvature would be at that scale, coated with a thin layer of light-sensitive chemical called *resist*. The resist plays the role of photographic film: where the electron beam hits it, its chemistry changes, and after development the exposed pattern becomes a stencil for the next chip-fabrication step.

The wafer sits on a wafer chuck — an aluminum-nitride ceramic plate with a thin layer of helium gas underneath it to conduct heat, plus a liquid coolant loop, plus 36 individually-controlled little thermoelectric heaters and coolers (think computer Peltier coolers, but precision-grade) to keep the wafer at exactly the right temperature to within 8 thousandths of a degree across its full 300 mm diameter.

Why so fussy about temperature? Because silicon expands when heated. A temperature change of one degree across a 300 mm wafer would move features by hundreds of nanometres — completely wrecking the alignment between this layer and the next. The whole chuck-and-coolant arrangement is essentially a tabletop precision oven, running cold and steady.

A piezoelectric stage moves the wafer back and forth under the column, scanning across the full 300 mm diameter in a serpentine pattern. The stage position is monitored by a laser interferometer (the Zygo ZMI) with a resolution finer than 0.1 nanometres — about the size of a single atom — so the system always knows exactly where on the wafer it is writing.

3.3 The beams: a million brushes

Each beam is responsible for writing patterns inside its own 20-micrometre-square tile. Within that tile, the beam can be deflected to any of about 700,000 distinct positions, depending on the chip’s resolution requirements. Each pixel takes about 3 nanoseconds to write. A full wafer’s pattern flows through in about 30 to 60 minutes per layer.

Different beams write different things. The whole pattern of the chip is sliced up into a million little patches, one per beam, and each beam gets a custom instruction stream telling it where to deflect, when to blank on, when to blank off.

Some of the beams aren't writing chip patterns at all. About one thousand of the million beams — a tenth of one percent — are dedicated *registration probes*. They scan over little reference marks (fiducials) etched onto the wafer, and the reflected electrons tell the control system exactly where the wafer is positioned, in real time. This lets the system correct for any tiny drift in the stage, the column, or the wafer chuck, with sub-nanometre precision.

3.4 The data path: through one wire vs. through many

Here is a subtle problem that turns out to dominate the cost of the system. A million beams, each switching at gigahertz speeds, generate an enormous amount of instruction-data. The raw stream — before any compression — is about 400 terabits per second. That is roughly *fifty thousand times* the average home internet connection.

You cannot send 400 terabits per second through copper wires. Copper has resistance, copper has heat, and routing copper into a cryogenic ultra-high-vacuum chamber is a thermal nightmare — every copper wire is a tiny heat leak from room temperature into the cold zone. (Imagine threading a hundred space heaters through a sleeping bag and expecting it to stay warm.)

The solution is to send the data through optical fibre. Photons carry data with zero resistance, zero conduction, and essentially zero heat leak. Fibres are made of glass, which is an excellent thermal insulator. The full data path uses 32 fibre trunks, each carrying 96 different wavelengths of light (a technique called dense wavelength-division multiplexing, the same trick that lets one undersea cable carry every transatlantic phone call simultaneously), each wavelength running at 200 gigabits per second.

That is 608 terabits per second of raw capacity, which after software compression carries the actual 50-terabit-per-second pattern stream comfortably with room to spare.

At the cold end, the optical signals are converted back to electrical signals by tiny photodiode chips made of indium phosphide, which work well at 77 K. The electrical signals then go into a layer of *cryo-CMOS* digital electronics — silicon chips specifically designed to operate at cryogenic temperature — that distribute the instructions to each of the million coils.

This is “internet through many wires” instead of “internet through one wire.” It is also the difference between an architecture that works and one that overheats.

3.5 The control system: the conductor of a million-piece orchestra

Coordinating a million independent beams in real time, with sub-nanometre precision, with all the various physical effects (thermal drift, wafer movement, source-emission variations) being measured and compensated continuously, is a serious software and control problem.

The architecture uses what's called a *multi-rate control hierarchy*. Think of it as the multi-tiered cam-timing of a complex engine: different things happen at different speeds.

- **Inner loop, 10,000 times per second:** each individual beam's deflection is updated based on its current pattern instruction and any in-flight registration corrections. This loop runs on small RISC-V processor cores embedded inside the cryo-CMOS chips, in the cold zone.

- **Middle loop, 1,000 times per second:** the blanker turns each beam on or off in synchrony with the pattern data.
- **Outer loop, 1 to 10 times per second:** the wafer stage moves, the column’s overall position is checked against the master laser interferometer, and broad corrections are pushed back into the inner loops.
- **Supervisory layer, real-time but humans can watch:** the warm-side computer that the operator interacts with, handles recipes, talks to the rest of the fab’s IT, logs everything.

The control engineering analogue is “building a Lego model while the entire room is rotating, the floor is heaving, and the table is being subtly stretched.” The multi-rate hierarchy lets each subsystem worry about only its own time scale while passing the relevant information up and down the stack.

The full software stack — pattern compiler that turns chip-design files into beam instructions, real-time firmware that runs on the cryo-CMOS cores, supervisory operating system that handles the fab integration — is also published openly. Every library and framework used is open-source. PostgreSQL for the calibration database. Next.js for the operator user interface. Tango Controls for the real-time orchestration. The Linux operating system. There are no proprietary blackboxes anywhere in the software chain.

4 The numbers that matter

Here are the headline numbers, with context for what they mean.

Quantity	Number	What this means
Beams per machine	1,000,000	A million electron beams in parallel
Time per pixel	3 ns	One pixel every three billionths of a second
Pattern data rate	50 Tbps	Roughly 6,000 4K Netflix streams
Resolution, normal mode	50–180 nm	About the size of a virus particle
Resolution, slow mode	down to 7 nm	About 30 silicon atoms wide
Throughput	8–17 wafers/hour	One wafer every 4–7 minutes
Capital cost per tool	\$37–42 M	Compare to \$200–400 M for EUV
Cold operating temperature	77 K (−196°C)	Liquid nitrogen temperature
Prototype cost	~\$600 k	Small enough for a university lab
Prototype timeline	18 months	One graduate-student PhD cycle

A few of these deserve a closer look.

What 50 nm feels like. Fifty nanometres is roughly the size of a single virus particle — say, an influenza virus. It is also roughly the smallest feature size on the chips that run a typical car’s anti-lock braking system, an industrial robot, an MRI machine, a satellite radio, or a CT scanner. About 80% of the world’s semiconductor demand by chip-count is in this size range or larger. It is *not* what the latest iPhone uses (those use 3 nm features) but it is what almost every other piece of electronics on Earth uses.

What 7 nm feels like. Seven nanometres is the smallest feature the machine can write, accessed by running each beam very slowly (and therefore the whole machine more slowly). That is about 30 silicon atoms across — *the actual atomic scale*. At this size, drawing a pattern is like painting

on a jigsaw puzzle where every piece is one atom of paint.

What “8 to 17 wafers per hour” means. A modern EUV scanner can do maybe 180 wafers per hour at high-volume nodes — about ten times faster. So this machine is not a high-volume rival to EUV. It’s a different category: low-to-mid-volume, high-flexibility, low-cost-of-entry. The throughput is comparable to a small-batch food production line versus a Tyson-scale factory.

What \$37 million sounds like. A new commercial airliner costs \$100 million. A modern mid-size MRI machine, \$3 million. A high-end industrial CNC mill, \$2 million. Thirty-seven million dollars is in the same range as a high-end DUV (deep-UV, the lithography generation before EUV) scanner, which is the tool most of the world’s chip fabs are actually still buying.

What “\$600,000 prototype” means. A well-funded university research group, a small national lab, or a Series A startup can afford a \$600,000 piece of equipment. EUV at \$400 million is a sovereign-level commitment. This is a startup-grade commitment.

5 Why “open” matters

A great many things in modern technology are technically remarkable but locked behind intellectual property and supply-chain bottlenecks. ASML’s EUV is the canonical example. Even if you, as a sovereign nation or a well-funded competitor, *wanted* to build your own EUV scanner, you would be unable to: thousands of patents block every step, every supplier has exclusive contracts with ASML, and your engineers would be welcomed in court more than in workshops.

This architecture deliberately walks the other direction.

5.1 What “open” means in practice

The Morin 2026 v5 package is released under the Creative Commons Attribution license for the documents and the MIT license for the software. *By publication, every idea in the package has become prior art*, which is a legal term meaning: no one, including the original author, can later patent it. The architecture is in the public domain. Forever. Any country, any university, any company, any individual can build it, modify it, sell it, fork it, improve it.

The author makes no patent claim on anything. There is no exclusive licensing. There is no NDA to sign. There is no “we’ll grant you the rights if you agree to terms” gate. The terms are: do what you want, give credit, ideally share back improvements.

5.2 The supply chain is genuinely plural

Open documents are necessary but not sufficient. You also need to be able to *buy* the parts. Every subsystem in this architecture has been specified with at least three independent vendors in at least two different countries. To pick a few examples:

- **Superconducting tape (YBCO):** SuperPower (USA), American Superconductor (USA), SuperOx (Russia/Japan), Furukawa (Japan).
- **Cryocoolers:** Cryomech (USA), Sumitomo (Japan), Brooks (USA).
- **Cryo-CMOS fabrication:** GlobalFoundries (USA/Singapore), TSMC (Taiwan), Imec (Belgium), Fraunhofer (Germany).

- **Optical fibres and transceivers:** Corning (USA), Furukawa (Japan), Sumitomo (Japan), plus a dozen Chinese suppliers.
- **Field emitters:** NSC (Canada), CEA-Leti (France), Imec (Belgium).
- **Laser interferometers:** Zygo (USA), Heidenhain (Germany), Renishaw (UK).

If the U.S. government tomorrow forbade the export of any single component, the machine could still be built using alternative suppliers — in most cases without leaving North America, and in all cases without leaving the OECD.

This is not by accident. ASML’s value proposition (and vulnerability) is that they are the *only* people who put the pieces together. This architecture’s value proposition is the opposite: there is no chokepoint, anywhere.

5.3 The geopolitical implications

The current global semiconductor supply is geographically concentrated to a degree that has alarmed every defence ministry on Earth. Roughly 70% of leading-edge wafers come out of Taiwan. Roughly 90% of the EUV scanners come out of one factory in the Netherlands. The most complex chip-design software comes from three U.S. companies. The most advanced packaging comes from a handful of locations in Korea and Taiwan. A war, a pandemic, a natural disaster, or a single regulatory decision can disrupt the entire global supply.

A multi-vendor, open-architecture mature-node tool does not solve this problem at the bleeding edge — it does not, by itself, let a sovereign nation make 3 nm Apple silicon. But it does offer something that has previously been unavailable: a way for any moderately-resourced nation (or large company, or consortium of universities) to make their own *mature-node* chips, in country, without external dependency. And mature-node chips are 80% of the world’s demand. They run the cars, the medical devices, the satellites, the power grids, the industrial automation. They are the chips that, when their supply is interrupted, the modern economy notices first.

Chips that don’t have to come from Taiwan or Texas. That’s the strategic prize.

6 What you could make with it

If a machine like this existed and was widely available, what would change?

6.1 Specialty chips for defence

Modern radar systems, secure communication systems, electronic warfare systems, encryption hardware, satellite payloads — these all need custom chips that are typically made in low volumes (hundreds to low thousands of units), where mask cost dominates everything. A mask-free tool would slash the cost of custom defence silicon by an order of magnitude, and let it be made in country without exposing the design to commercial foundries.

6.2 Medical implants and biocompatible chips

Cochlear implants, retinal prostheses, neural interfaces, pacemakers, implantable insulin pumps. The total worldwide demand for each of these is millions of units per year — not enough to justify a \$50M mask set. Currently each project either pays the mask premium or settles for off-the-shelf

chips that aren't well-suited. A direct-write tool would let medical-device companies do custom designs at modest cost.

6.3 Specialised AI accelerators for narrow uses

Modern general-purpose AI chips (NVIDIA H100s, Google TPUs) are designed for fungible large-volume use. There are many AI applications — edge inference for industrial vision systems, on-device speech recognition for hearing aids, low-power neural processors for ag-tech sensors — where a custom narrow-purpose chip would be vastly more efficient than a general-purpose chip. Today they aren't economically viable because of mask cost. Tomorrow they could be.

6.4 Satellite electronics

Radiation-hardened space electronics are a low-volume specialty market currently served by a small number of suppliers at very high markup. Mask-free direct-write would let satellite makers iterate designs much faster, much cheaper, and respond to mission-specific requirements without the year-long mask-set cycle.

6.5 Industrial automation

Motor controllers, sensor processors, power-electronics chips for factory floors and renewable-energy installations. Mature-node territory. Currently fine for big consumer markets but expensive for niche applications. Could open up a long tail of custom industrial silicon.

6.6 University research

A \$600k university lab tool, built openly, scales down to the *capability* of full-scale multi-beam lithography. Graduate students doing nanoelectronics research could iterate on real silicon devices with day-scale turnaround instead of month-scale. Materials scientists could explore new device architectures without the mask-cost penalty. The biological-inspired computing community, the spintronics community, the photonic-computing community could all build experimental devices that today they cannot afford to prototype.

6.7 Multi-material chips

Because the tool writes directly, without a mask, it can be used in process flows that combine multiple materials in ways that conventional lithography cannot. Silicon plus carbon nanotube logic. Silicon plus gallium nitride power transistors. Silicon plus diamond heat spreaders. Silicon plus exotic 2D materials like molybdenum disulfide. Each of these material combinations needs custom mask sets per material, and integrating them is expensive. Mask-free direct-write opens the door to heterogeneously integrated chips at moderate cost.

This last category — multi-material chips — may be the most strategically interesting. The end of Moore's Law (the historical pattern of transistors getting smaller every 18 months) means that future performance gains come from *integration* rather than from feature shrink. Direct-write tools are uniquely suited to this future.

7 How it gets built

The development of any complex piece of capital equipment proceeds in stages. The v5 architecture lays out a five-stage path.

7.1 Stage A: the small prototype

Cost: ~\$600,000. **Timeline:** 18 months. **Beams:** 1 (in this stage, only a single beam is used).

The goal of Stage A is to validate the fundamental physics of the architecture with one beam. Specifically:

1. Does a YBCO superconducting deflection coil at 77 K bend an electron beam as predicted? (Closes the basic deflection-kinematics question.)
2. Does a YBCO Meissner shield block magnetic crosstalk by the predicted 80 decibels or more? (Closes the magnetic-isolation question.)
3. Does the cryo-CMOS digital-to-analog converter, driving the coil through an optical fibre from a warm-side controller, work? (Closes the data-path question.)
4. Does the backscatter-electron registration loop close with sub-2-nm accuracy? (Closes the registration-loop question.)
5. Does cryosorption capture resist outgassing as predicted? (Closes the vacuum-management question.)
6. Does the wafer-chuck thermal management work at scaled flux? (Closes the wafer-thermal question.)

If all six tests pass within their design margins, the v4 architecture is empirically validated at single-beam scale. Stage A's bill of materials is published in the open package; every component is off-the-shelf from a known commercial vendor.

This is the stage that is buildable *today*, by any well-resourced university, national laboratory, or Series A semiconductor startup.

7.2 Stage B: the few-beam prototype

Cost: ~\$5 million. **Timeline:** 24 months from Stage A completion. **Beams:** 16 to 64.

Stage B builds a small array (say, an 8×8 grid of beams) inside a production-grade cold column. This validates cross-beam interactions, the photonic data path at scale, source closed-loop control, and the multi-beam registration system. This is the critical de-risking stage: if Stage B works, scaling further is engineering, not science.

7.3 Stages C and D: scaling and production

Stage C cost: \$10–15 million. **Timeline:** 24 months. **Beams:** 1,000 to 10,000. First exposures of real wafers.

Stage D cost: \$30–40 million. **Timeline:** further 12–24 months. **Beams:** 100,000 to 1,000,000. Integration with standard wafer-processing line. This is the first production tool.

Total program cost from start to first production tool: \$45–60 million. **Total timeline:** about 4–5 years.

For context: ASML’s EUV program ran for over twenty years, cost tens of billions of dollars, and required the cooperation of nearly every major chip company on Earth. A \$50M / 5-year program is a different order of magnitude entirely. It is small enough to fit inside a single sovereign defence appropriation, a single large-university research budget, or a single Series B/C startup raise.

7.4 The open question

Who builds the first one?

There are perhaps a dozen places on Earth that could plausibly fund Stage A out of existing budgets and have the in-house expertise to execute it: defence research agencies (DARPA, DRDC, DRDO, others), large semiconductor research consortia (Imec, SEMATECH-equivalent), national labs (Sandia, LLNL, Berkeley), and several semiconductor-focused universities. There are similarly a handful of startup founders who could raise \$5M for Stage B if Stage A succeeds.

The architecture is in the public domain. The path is documented down to the bill of materials. The only thing missing is a builder. The author’s hope is that publication-as-prior-art removes the “who owns this?” hesitation that has historically blocked similar efforts.

8 What’s NOT promised

Honest engineering documents are as valuable for what they decline to claim as for what they do claim. Here is a clear-eyed list of what this architecture is *not* going to do.

8.1 It will not replace EUV at the cutting edge

A modern EUV scanner can produce roughly 180 wafers per hour at the 3-nanometre node, with the resolution and overlay accuracy needed for billion-transistor Apple and NVIDIA chips. This architecture, at its fastest, does maybe 17 wafers per hour at the 50-nanometre node. The two machines are not in the same category. EUV will continue to make iPhones; this machine will make almost everything else.

8.2 It will not make 3 nm chips at production speed

The architecture *can* access feature sizes down to 7 nm — but only by slowing down the per-beam current dramatically, which slows the throughput correspondingly. Making 7 nm chips with this tool would be appropriate for prototyping, R&D, low-volume specialty work, or niche custom chips — not for high-volume consumer products.

8.3 It will not be cheap to develop

The Stage A prototype is \$600k. The full production tool is \$40M to develop. The total program through first production tool is \$50–60M. That is cheap *compared to EUV* (twenty thousand times cheaper) but it is not cheap in absolute terms. This is a multi-year, multi-team, multi-vendor engineering program — not a garage project.

8.4 There will be surprises

Every precision-instrument program runs into process-development surprises that delay it. The YBCO 20-micrometre patterning has not been done at full production scale. The cryo-CMOS DAC at this channel count is a new chip design. The MEMS source array needs to scale from current 10,000-tip arrays to a million tips. None of these are *physics-blocking*, but each will have its own process-development arc, and each will take longer than the optimistic Gantt chart suggests. The program needs realistic engineering management, contingency budgets, and patience.

8.5 It is theoretical engineering, not built hardware

The v5 package is a *complete design package*, not a delivered product. The physics has been simulated end-to-end and the engineering has been spec'd subsystem-by-subsystem with vendor candidates. But the integrated machine does not yet exist. The package is an invitation, with a credible road map and an open license, for someone to build it.

9 What this means

Let me close with what this architecture means in the bigger picture.

For thirty years, advanced semiconductor manufacturing has been concentrating into fewer hands, fewer factories, fewer countries. The reasons are partly economic (chip fabs have enormous economies of scale), partly technological (each generation is harder than the last), and partly geopolitical (the value of being the indispensable supplier is enormous). The result is that the modern world's most important physical infrastructure — the basis of every computing device, every car, every aircraft, every medical scanner, every weapon system — depends on a handful of factories on a handful of islands, on the political stability of a handful of regimes, on the goodwill of a handful of companies.

This architecture does not undo that. It would be foolish to claim it does. The leading edge — TSMC's 2-nm process, Apple's M-series silicon, NVIDIA's largest GPUs — remains a place where EUV is the only game in town for the foreseeable future.

But it offers something new for the *rest* of the chip world. It offers the possibility that any moderately developed country, any sufficiently funded university, any sufficiently determined startup can build their own custom chips, at mature nodes, in moderate volumes, without external dependency. It offers the possibility that the supply chain for the world's most important physical infrastructure is no longer a single-point-of-failure, that there are multiple paths to chips, that no one company or country can hold the rest hostage.

It also offers a particular philosophical statement. The author of this architecture is one person, working alone, with no external funding, on a commodity Linux laptop. The supporting simulations were run on a desktop computer. The author makes no patent claim, takes no licensing fee, and releases the entire work under the most permissive licenses available. This is a deliberate counter-example to the dominant model in semiconductor manufacturing — and a demonstration that even in the most capital-intensive, most secretive, most monopolistic industry on Earth, an individual using modern tools can produce a complete, credible, openly-released alternative architecture.

Whether anyone picks it up and builds it is the open question. The architecture is closed. The math is honest. The supply chain is plural. The patent landscape is clean. The build is hard but possible.

The next move is somebody else's.

Robert Gerald Morin, May 2026, Edmonton, Alberta, Canada. Released under CC BY 4.0. The full technical preprint, all simulation code, all subsystem specifications, and all software reference implementations are available in the open-source package `morin_2026_multibeam_ebl_open` at the upstream repository. Build it, fork it, improve it, share it back. Architecture compounds in value when it is shared.